

An Independent Recursive Geometric Derivation of π Using Inscribed Polygons and Its High-Precision Computational Validation

Self-Directed Mathematical(Grade 10) & Computational(University) Study

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Abstract

This document presents an independent recursive derivation of an infinite nested radical formulation for π , originally conceptualized during Grade 10 secondary school studies without prior knowledge of calculus or Viète's formula. Using iterative Pythagorean geometry on inscribed polygons ($N = 4, 8, 16, 32, \dots, 2^{n+2}$), explicit recurrence relations were formulated. Due to the lacking of programming tools at the time of this derivation, the algorithmic convergence was later computationally validated during university studies using MATLAB high-precision symbolic arithmetic ('digits(100)'). The algorithm exhibits exponential convergence, correctly evaluating π to 31 decimal places at $n = 50$, seamlessly bridging elementary geometry with arbitrary-precision numerical computing.

1 Introduction

The mathematical constant π represents the fundamental ratio of a circle's circumference C to its diameter D . While historically evaluated through infinite series and calculus-based trigonometric identities, this work presents a purely geometric reconstruction derived from first principles.

This formulation originated during Grade 10 secondary school studies as an independent geometric exploration. By repeatedly doubling the sides of an inscribed square ($N = 4$) and applying Pythagorean relations, a recursive sequence of nested square roots was established. However, because programming and high-precision computational environments were unavailable during high school, the numerical verification of this recursive limit remained unrealized.

Later, during undergraduate engineering studies, the algorithm was implemented in MATLAB using high-precision symbolic computation ('digits(100)'). This paper documents both the original high school geometric derivation and its subsequent university-level computational validation, demonstrating the rapid numerical convergence of the sequence.

2 Initial Geometric Framework: Inscribed Square ($N = 4$)

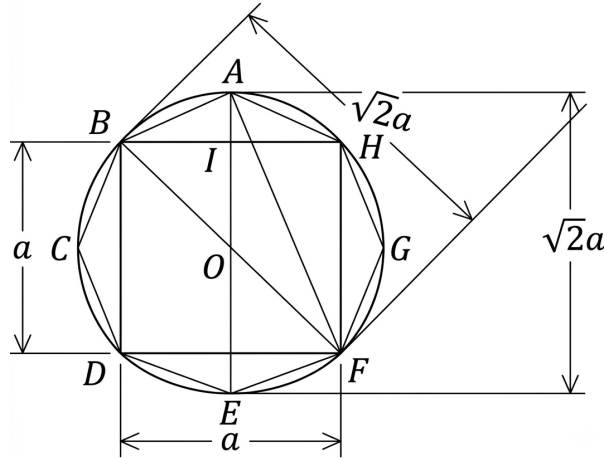


Figure 1: Inscribed Square ($N = 4$) and 8-Polygon Construction

Consider a circle centered at O . Inscribe a square $BHFD$ within the circle with side length a :

$$BH = FH = DF = BD = a \quad (1)$$

2.1 Diameter and Radius

The diagonal of the inscribed square corresponds directly to the diameter D of the circle:

$$\text{Diagonal } BF = \text{Diameter } D = \sqrt{a^2 + a^2} = \sqrt{2}a \quad (2)$$

Since $D = 2R$, the radius R is:

$$R = AO = BO = FO = \frac{a}{\sqrt{2}} \quad (3)$$

3 Step-by-Step Iterative Derivations

3.1 First Iteration: Inscribed 8-Polygon ($N = 8$)

From figure 1, let I be the midpoint of chord BD , so $BI = \frac{a}{2}$. The segment OI represents the apothem:

$$OI = \frac{a}{2} \quad (4)$$

The radial segment AI along axis AO is:

$$AI = AO - OI = R - \frac{a}{2} = \frac{a}{\sqrt{2}} - \frac{a}{2} = a \left(\frac{1}{\sqrt{2}} - \frac{1}{2} \right) \quad (5)$$

In the right-angled triangle $\triangle ABI$:

$$\begin{aligned}
AB^2 &= AI^2 + BI^2 \\
&= \left(\frac{a}{\sqrt{2}} - \frac{a}{2} \right)^2 + \left(\frac{a}{2} \right)^2 \\
&= \left(\frac{a^2}{2} - \frac{a^2}{\sqrt{2}} + \frac{a^2}{4} \right) + \frac{a^2}{4} \\
&= a^2 \left(1 - \frac{1}{\sqrt{2}} \right)
\end{aligned} \tag{6}$$

Taking the positive root yields the side length of the 8-gon (AB):

$$AB = a\sqrt{1 - \frac{1}{\sqrt{2}}} \quad (7)$$

3.1.1 Perimeter-to-Diameter Ratio for $N = 8$

The total perimeter $P_8 = 8 \cdot AB = 4 \times 2 \times a\sqrt{1 - \frac{1}{\sqrt{2}}}$. The perimeter-to-diameter ratio is:

$$\frac{P_8}{D} = \frac{4 \times 2 \times a \sqrt{1 - \frac{1}{\sqrt{2}}}}{\sqrt{2}a} = \frac{2^{1+2}}{\sqrt{2}} \sqrt{1 - \frac{1}{\sqrt{2}}} \approx 3.06146 \dots \quad (8)$$

3.2 Second Iteration: Inscribed 16-Polygon ($N = 16$)

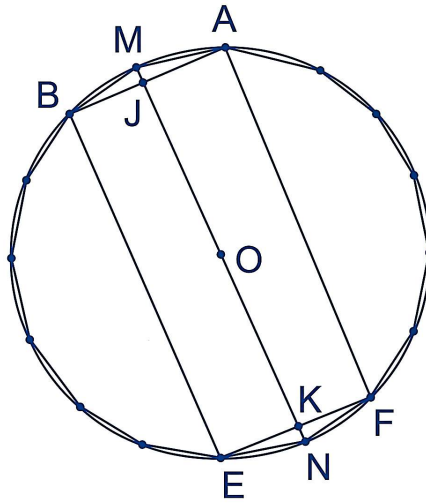


Figure 2: Subdivision into 16-Polygon and Mid-chord Analysis

To progress to the 16-gon, we derive the perpendicular segment AF via $\triangle AEF$ from figure 1, where $EF = a\sqrt{1 - \frac{1}{\sqrt{2}}}$ and $AE = AO + OE = \frac{a}{\sqrt{2}} + \frac{a}{\sqrt{2}} = \sqrt{2}a$:

$$\begin{aligned} AF^2 &= AE^2 + EF^2 \\ &= (\sqrt{2}a)^2 - a^2 \left(1 - \frac{1}{\sqrt{2}}\right) = 2a^2 - a^2 \left(1 - \frac{1}{\sqrt{2}}\right) \\ &= a^2 + \frac{a^2}{\sqrt{2}} = a^2 \left(1 + \frac{1}{\sqrt{2}}\right) \end{aligned} \quad (9)$$

Thus, in figure 2 segment AF (and JK) is:

$$AF = JK = a\sqrt{1 + \frac{1}{\sqrt{2}}} \quad (10)$$

Defining J as the midpoint of chord AB , segment MJ is:

$$\begin{aligned}
 MJ &= MO - OJ = MO - \frac{JK}{2} = R - \frac{a}{2} \sqrt{1 + \frac{1}{\sqrt{2}}} \\
 &= \frac{a}{\sqrt{2}} - \frac{a}{2} \sqrt{1 + \frac{1}{\sqrt{2}}}
 \end{aligned} \tag{11}$$

Applying the Pythagorean theorem in $\triangle BMJ$ with $BJ = \frac{AB}{2} = \frac{a}{2}\sqrt{1 - \frac{1}{\sqrt{2}}}$:

$$\begin{aligned}
BM^2 &= MJ^2 + BJ^2 \\
&= \left(\frac{a}{\sqrt{2}} - \frac{a}{2}\sqrt{1 + \frac{1}{\sqrt{2}}} \right)^2 + \left(\frac{a}{2}\sqrt{1 - \frac{1}{\sqrt{2}}} \right)^2 \\
&= \frac{a^2}{2} - \frac{a^2}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}} + \frac{a^2}{4} \left(1 + \frac{1}{\sqrt{2}} \right) + \frac{a^2}{4} \left(1 - \frac{1}{\sqrt{2}} \right) \\
&= a^2 - \frac{a^2}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}} = a^2 \left(1 - \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}} \right)
\end{aligned} \tag{12}$$

Thus, the side length of the 16-gon (BM) is:

$$BM = a\sqrt{1 - \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}}} \tag{13}$$

3.2.1 Perimeter-to-Diameter Ratio for $N = 16$

With $P_{16} = 16 \cdot BM = 4 \times 2 \times 2 \times BM$:

$$\frac{P_{16}}{D} = \frac{4 \times 2 \times 2 \times a\sqrt{1 - \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}}}}{\sqrt{2}a} = \frac{2^{2+2}}{\sqrt{2}}\sqrt{1 - \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}}} \approx 3.12144 \dots \tag{14}$$

3.3 Third Iteration: Inscribed 32-Polygon ($N = 32$)

Extending the exact geometric construction using radius and diameter of the circle, also the side length and the half-side length of the 16-polygon yields the square of the side length for a 32-polygon:

$$\begin{aligned}
\text{Side Length}^2 &= \left[\frac{a}{\sqrt{2}} - \sqrt{(\sqrt{2}a)^2 - \left(a\sqrt{1 - \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}}} \right)^2} \right]^2 + \left(\frac{a}{2}\sqrt{1 - \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}}} \right)^2 \\
&= \left(\frac{a}{\sqrt{2}} - \frac{a}{2}\sqrt{1 + \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}}} \right)^2 + \left(\frac{a}{2}\sqrt{1 - \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}}} \right)^2 \\
&= a^2 \left(1 - \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}}} \right)
\end{aligned} \tag{15}$$

Length for the 32-gon (s_{32}) is:

$$s_{32} = a\sqrt{1 - \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}}}} \tag{16}$$

3.3.1 Perimeter-to-Diameter Ratio for $N = 32$

With total perimeter $P_{32} = 32 \cdot s_{32} = 32 \times a \cdot s_{32}$:

$$\frac{P_{32}}{D} = \frac{32 \times a\sqrt{1 - \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}}}}}{\sqrt{2}a} = \frac{2^{3+2}}{\sqrt{2}}\sqrt{1 - \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}\sqrt{1 + \frac{1}{\sqrt{2}}}}} \approx 3.13654 \dots \tag{17}$$

4 Generalization and Analytical Limit for π

For the n -th iteration step corresponding to a regular polygon with $N = 2^{n+2}$ sides (where $n \geq 1$ denotes the iteration order, containing n nested square roots):

The perimeter-to-diameter ratio $\frac{P_n}{D}$ is expressed as:

$$\frac{P_n}{D} = \frac{2^{n+2}}{\sqrt{2}} \underbrace{\sqrt{1 - \frac{1}{\sqrt{2}} \sqrt{1 + \frac{1}{\sqrt{2}} \sqrt{1 + \frac{1}{\sqrt{2}} \cdots \sqrt{1 + \frac{1}{\sqrt{2}}}}}}}_{n \text{ nested square roots}} \quad (18)$$

Taking the limit as $n \rightarrow \infty$, the polygon perimeter P_n strictly converges to the circle's circumference C , yielding the infinite limit for π :

$$\pi = \lim_{n \rightarrow \infty} \frac{2^{n+2}}{\sqrt{2}} \sqrt{1 - \frac{1}{\sqrt{2}} \sqrt{1 + \frac{1}{\sqrt{2}} \sqrt{1 + \frac{1}{\sqrt{2}} \cdots \frac{1}{\sqrt{2}} \sqrt{1 + \frac{1}{\sqrt{2}}}}}} \quad (19)$$

5 Computational Validation via MATLAB

To test the numerical convergence speed of the derived nested radical formula, the recursive loop was implemented in MATLAB using Symbolic Math Toolbox with high-precision variable arithmetic ('digits(100)').

5.1 Theoretical Error Behavior

The asymptotic convergence rate of the recursive algorithm can be rigorously derived using the Taylor expansion of the inscribed polygon's perimeter. For a regular N -gon inscribed within a circle of diameter D , the perimeter-to-diameter ratio π_N is given by:

$$\pi_N = \frac{P_N}{D} = N \sin\left(\frac{\pi}{N}\right). \quad (20)$$

Since the polygon is inscribed, $\pi_N < \pi$ for all finite N , converging to π as $N \rightarrow \infty$. Applying the standard Taylor series expansion for sine:

$$\sin x = x - \frac{x^3}{6} + O(x^5) \quad (21)$$

Setting $x = \frac{\pi}{N}$ yields:

$$\begin{aligned} \pi_N &= N \left[\frac{\pi}{N} - \frac{1}{6} \left(\frac{\pi}{N} \right)^3 + O\left(\frac{1}{N^5} \right) \right] \\ &= \pi - \frac{\pi^3}{6N^2} + O\left(\frac{1}{N^4} \right). \end{aligned} \quad (22)$$

Rearranging terms, the leading-order geometric approximation error is:

$$\pi - \pi_N \sim \frac{\pi^3}{6N^2} \quad (N \rightarrow \infty). \quad (23)$$

In this recursive construction, the number of polygon sides doubles at each iteration according to $N = 2^{n+2}$. Substituting N into the error expression gives:

$$\pi - \pi_n \sim \frac{\pi^3}{6(2^{n+2})^2} = \frac{\pi^3}{96} \cdot 4^{-n}. \quad (24)$$

Defining $E_n = |\pi - \pi_n|$, the asymptotic error behavior is bounded by:

$$E_n = O(4^{-n}) \quad (25)$$

This confirms that each iteration reduces the leading geometric approximation error by a factor of approximately four, establishing the theoretical foundation for the algorithm's rapid convergence.

5.2 MATLAB Implementation Code

The algorithm is implemented in MATLAB using the Symbolic Math Toolbox with 100-digit precision ('digits(100)'). While the inner 'for' loop executes $m = 49$ recursive iterations, the final assignment for y introduces an additional outer square root layer. Consequently, the total depth of nested radicals evaluated is $n = 50$, corresponding to an inscribed polygon of $N = 2^{50+2} = 2^{52}$ sides.

```
clear; clc;
digits(100);

% Initial values for recursive inner terms
x = sym(1);
m = 49;

% Recursive loop generating 49 inner nested square root layers
for i = 1:m
    x = sqrt(sym(1) + (x/sqrt(sym(2))));
end

% Final expression adds an extra outer radical layer
% Total polygon sides: N = 2^(50+2) = 2^52
y = (2^(m + sym(5)/2)) * sqrt(sym(1) - (x/sqrt(sym(2))));

fprintf('Calculated_pi:\n%s\n', char(vpa(y, 60)));
fprintf('Actual_pi:\n%s\n', char(vpa(sym(pi), 60)));
```

Listing 1: High-precision MATLAB evaluation for $N = 2^{n+2} = 2^{m+3}$ inscribed polygon

5.3 Convergence Analysis and High-Precision Results

Executing the algorithm for a polygon of sides $N = 2^{50+2} = 2^{52}$ produces a numerical approximation that agrees with π to ****31 decimal places****:

Calculated $\pi = \mathbf{3.141592653589793238462643383279248096270709244061\dots}$

Actual $\pi = \mathbf{3.141592653589793238462643383279502884197169399375\dots}$

The geometric sequence exhibits rapid numerical convergence as the number of polygon sides is repeatedly doubled, with $N = 2^{n+2}$. The divergence observed beyond the 31st decimal place at $n = 50$ is primarily associated with subtractive cancellation in the evaluation of the term $\left(1 - \frac{x}{\sqrt{2}}\right)$ using high-precision variable-precision arithmetic (VPA).

Specifically, as n increases, the quantity $\frac{x}{\sqrt{2}}$ approaches 1, causing $\left(1 - \frac{x}{\sqrt{2}}\right)$ to become increasingly small. Consequently, the subtraction involves two nearly equal high-precision quantities, making the final result increasingly sensitive to finite-precision effects. This sensitivity is further amplified by the large scaling factor $2^{m+5/2} \approx 3.18 \times 10^{15}$ at $m = 49$.

6 Conclusion

This work demonstrates a complete journey from theoretical derivation to computational validation. The recursive geometric formulation, originally conceived during Grade 10 secondary school studies

through first-principles Pythagorean geometry, was successfully implemented and validated during university studies. The algorithm's ability to compute π accurately to 31 decimal places using MATLAB high-precision arithmetic confirms both the analytical rigor of the original geometric proof and its rapid algorithmic convergence.